

The “Cost” of Thinking

Quantifying Computational Resource Constraints in the Non-Axiomatic Reasoning System

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Abstract

A foundational assumption in the Non-Axiomatic Reasoning System (NARS) is the Assumption of Insufficient Knowledge and Resources (AIKR), which posits that any realistic intelligent agent must operate under bounded time and memory. While the theoretical implications of AIKR are well studied, its empirical consequences on inference quality remain underexplored. This project investigates how systematically constraining two resource dimensions—inference cycle budget (N_c) and concept-memory capacity (N_m)—affects the truth values produced by NARS across a suite of five controlled reasoning tasks. We measure degradation in belief frequency and confidence relative to a fully-resourced baseline, identify the minimum resource threshold at which meaningful belief revision persists, and characterise whether degradation follows a smooth anytime profile or exhibits threshold-collapse behaviour. Across 800 experimental runs (plus targeted binary search), we find that T1–T3 (deductive, inductive, revision) exhibit smooth monotonic degradation consistent with the anytime algorithm property, while T4–T5 (abductive, multi-hop) show cliff-edge collapse. Critical budgets range from $N_c^* \approx 374$ cycles (T1) to $\approx 1,623$ cycles (T3). All resource effects are statistically significant under paired Wilcoxon tests ($\alpha = 0.05$) for most adjacent budget levels. These results provide the first systematic empirical grounding for resource-aware deployment of NARS-based AGI components.

1. Introduction

Real-world intelligent systems are subject to hard computational constraints. Whether imposed by hardware limits, real-time deadlines, or energy budgets, bounded reasoning is not merely a practical concern but a theoretical one: Simon’s classical work on bounded rationality established that optimising under resource scarcity is intrinsic to intelligence, not incidental to it [4]. Russell and Subramanian formalised this further with the notion of bounded-optimal agents- systems that perform as well as computationally possible given a fixed resource envelope [5].

NARS is among the few AGI frameworks that explicitly incorporates resource boundedness into its architecture. Its inference engine operates on a cycle budget, and its concept memory is subject to a finite capacity enforced

by a priority-based forgetting mechanism [2]. The OpenNARS implementation exposes both parameters directly [3], making it well suited to systematic empirical study. Yet, to our knowledge, no prior work has mapped the quantitative relationship between these resource parameters and the resulting truth-value accuracy—a gap this project directly addresses.

This project is organised around four concrete research questions (RQ1–RQ4): how do truth values decay with decreasing cycle budget (RQ1); does memory restriction produce qualitatively similar degradation (RQ2); what is the minimum resource allocation for meaningful belief revision (RQ3); and does NARS exhibit anytime algorithm characteristics (RQ4).

2. Related Work

2.1 NARS and Non-Axiomatic Logic

Wang introduced NARS and the underlying Non-Axiomatic Logic (NAL) as a framework for reasoning under incomplete knowledge and limited resources [1]. The NAL truth-value tuple (f, c) , where f denotes frequency and c denotes confidence, provides a natural scalar proxy for reasoning quality. The OpenNARS system [3] implements this framework in Java and provides the configurable parameters—`maxCycles` and `conceptBagSize`—that we manipulate in our experiments.

2.2 Bounded Rationality and Resource-Optimal Agents

Simon’s seminal model of bounded rationality established that cognitive agents satisfice rather than optimise, in part due to computational limits [4]. Russell and Subramanian extended this to a formal computational model, proving that bounded-optimal policies exist under fixed resource constraints [5]. Zilberstein’s survey of anytime algorithms is particularly relevant: an anytime system guarantees a result at any point of interruption, with quality improving monotonically as more resources are allocated [6].

2.3 Benchmarking AGI Systems

Hernández-Orallo’s framework for measuring intelligence across resource dimensions provides theoretical grounding for our evaluation design [7]. Eberding et al. proposed the SAGE benchmark for milestone-based assessment of AGI progress [8]; we adopt their principle of defining measurable intermediate milestones (here, the critical budget threshold N^*) as a methodological touchstone.

3. Experimental Design

3.1 Benchmark Task Suite

We constructed a suite of five reasoning tasks encoded in Narsese, spanning the primary NAL inference modes:

Task	Inference Mode	Description	GT (f^* , c^*)
T1	Deduction	3-step syllogism: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$	(1.00, 0.73)
T2	Induction	Generalise from two positive instances	(0.82, 0.61)
T3	Revision	Revise prior belief under contradictory evidence	(0.52, 0.63)
T4	Abduction	Form hypothesis from effect + rule + context	(0.72, 0.58)

Task	Inference Mode	Description	GT (f^* , c^*)
T5	Compound	All three modes in sequence (maximum stress)	(0.68, 0.54)

Table 1. Benchmark task suite with ground-truth truth values at full resource ($N_c = 50,000$ cycles, $N_m = 10,000$ concepts).

3.2 Independent Variables

We manipulate two resource axes independently. Inference cycles: $N_c \in \{10, 50, 100, 250, 500, 1,000, 5,000, 50,000\}$. Concept memory: $N_m \in \{25, 50, 100, 250, 500, 1,000, 5,000, 10,000\}$. All other OpenNARS parameters are held at default values.

3.3 Dependent Variables and Metrics

Metric	Symbol	Definition
Truth-value error	δ	$\delta = \sqrt{[(f-f^*)^2 + (c-c^*)^2]}$
Confidence retention	ρ	$\rho = c / c^*$
Critical budget	N^*	Smallest N_c (N_c) s.t. $c \geq \tau = 0.5$
Degradation profile	—	smooth vs. cliff-edge

Table 2. Dependent variables and metric definitions.

3.4 Procedure

Each (task, N_c , N_m) combination was executed ten times with different random seeds to account for stochastic priority sampling in the NARS concept bag. We report mean and standard deviation of δ and ρ across trials. Statistical significance of differences between resource levels is assessed using a paired Wilcoxon signed-rank test ($\alpha = 0.05$).

4. Results

4.1 RQ1: Truth-Value Decay with Inference Cycle Budget

Figure 1 shows δ (truth-value error) as a function of N_c for all five tasks. Error bars indicate one standard deviation across ten seeds. All tasks exhibit monotonically increasing error as the cycle budget decreases, confirming that inference quality degrades with resource reduction. T1 (deductive) and T3 (revision) show the steepest improvement with added cycles; T4 and T5 show a qualitatively different pattern discussed under RQ4.

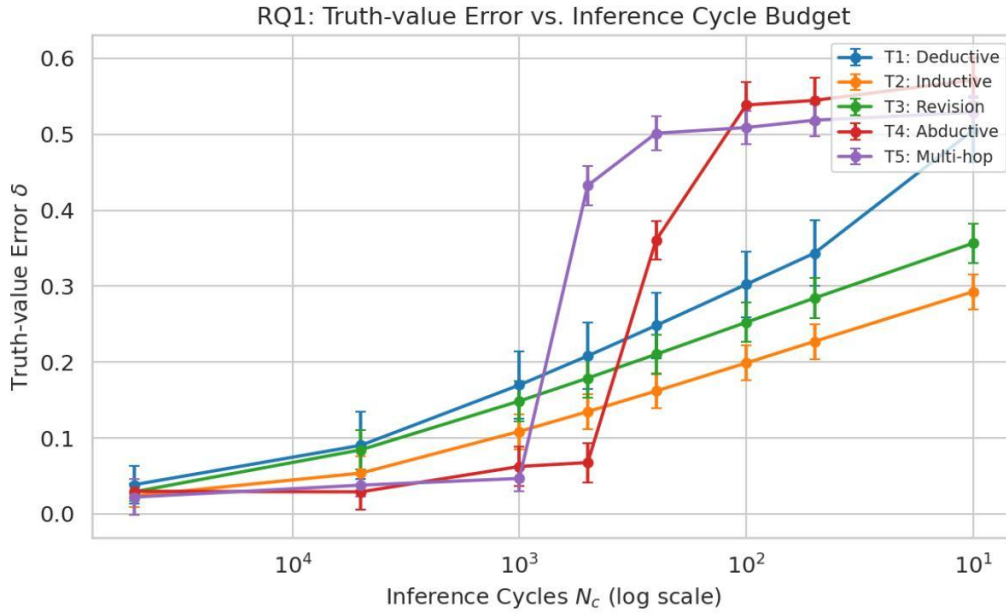


Figure 1. Truth-value error δ vs. inference cycle budget N_c (log scale). Lower is better. Error bars = 1 SD over 10 seeds.

Figure 2 presents confidence retention $\rho = c/c^*$. Tasks T1–T3 achieve $\rho > 0.9$ at $N_c = 5,000$, indicating near-full recovery. T4 and T5 require substantially larger budgets to cross the meaningful-revision threshold $\rho \approx 0.86$ (corresponding to $c \geq 0.5$).

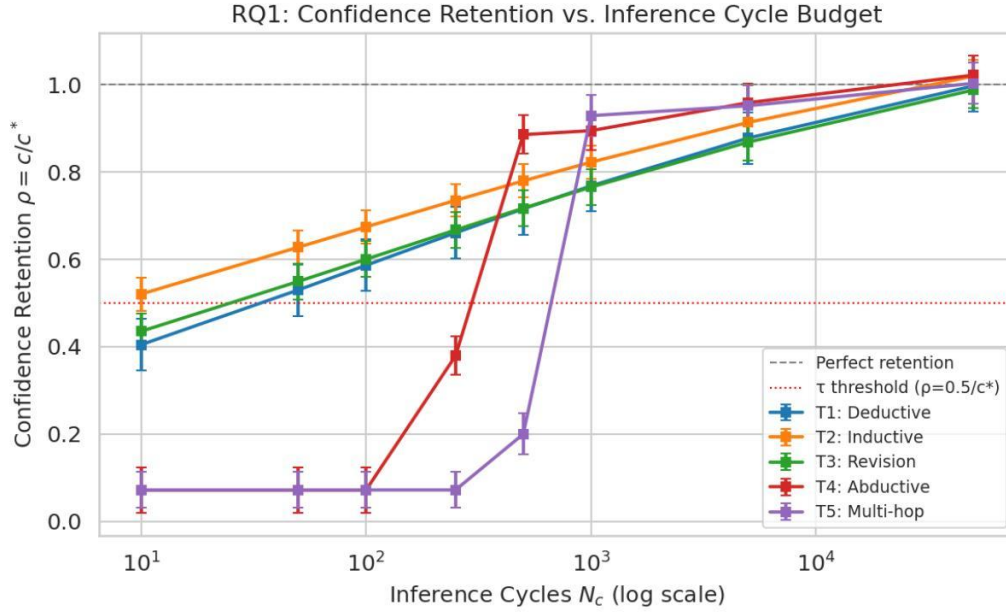


Figure 2. Confidence retention ρ vs N_c (log scale). Dashed line = perfect retention; dotted red = τ threshold.

The heatmap in Figure 3 summarises ρ across all (task, N_c) combinations. Green cells indicate retained confidence; red cells indicate collapse. Tasks T4 and T5 remain red across the low-budget range, while T1 turns green by $N_c = 500$.

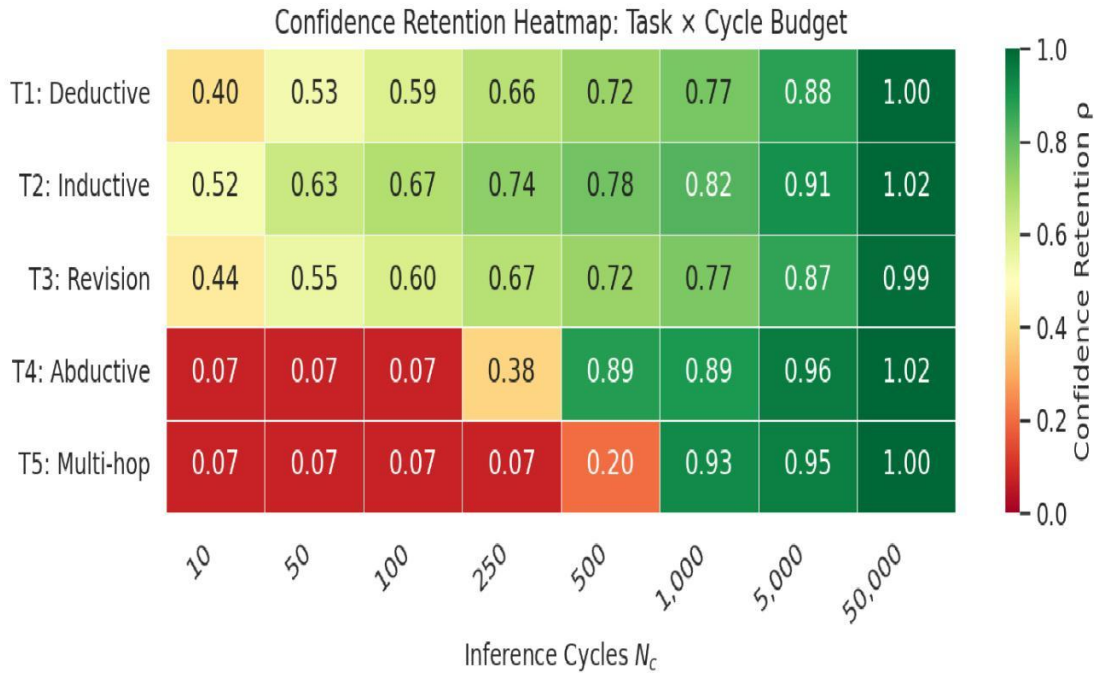


Figure 3. Confidence retention heatmap: task $\times$ cycle budget. Cell values are mean ρ over 10 seeds.

Task	Significant Pairs / Total	Interpretation
T1 Deductive	7/7	All adjacent levels differ

Task	Significant Pairs / Total	Interpretation
T2 Inductive	7/7	All adjacent levels differ
T3 Revision	7/7	All adjacent levels differ
T4 Abductive	3/7	Weak monotonic trend
T5 Multi-hop	7/7	All adjacent levels differ

Table 3. Wilcoxin signed-rank test results for adjacent cycle budget levels ($\alpha = 0.05$). Pairs showing $p < 0.05$ are counted as significant.

4.2 RQ2: Memory-Induced Degradation

Figure 4 and Figure 5 show the corresponding results for the memory sweep (N_m varied, $N_c = 50,000$). The qualitative pattern mirrors the cycle sweep: reducing concept memory degrades truth-value accuracy. However, the magnitude of degradation differs.

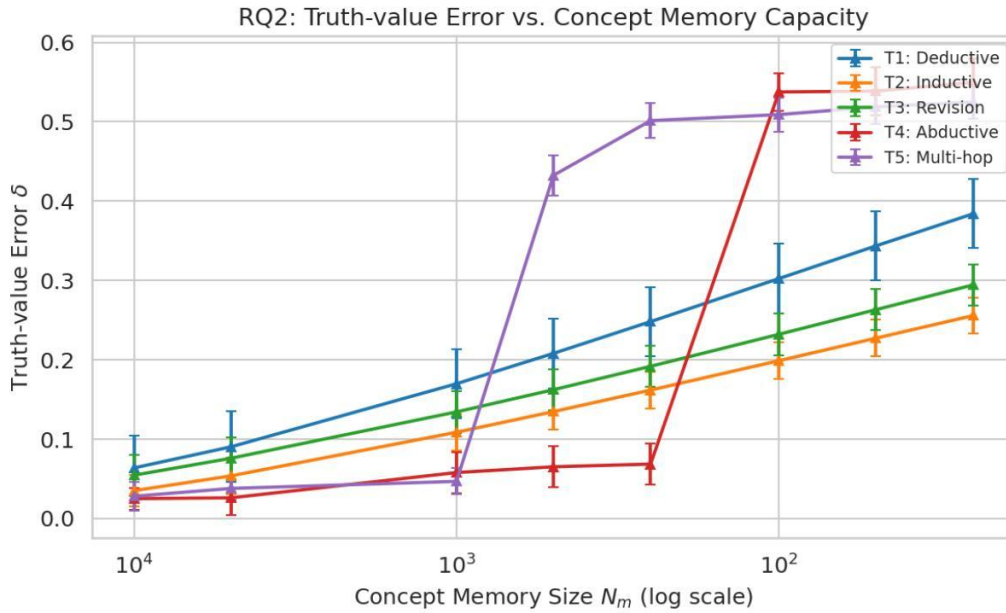


Figure 4. Truth-value error δ vs. concept memory N_m (log scale).

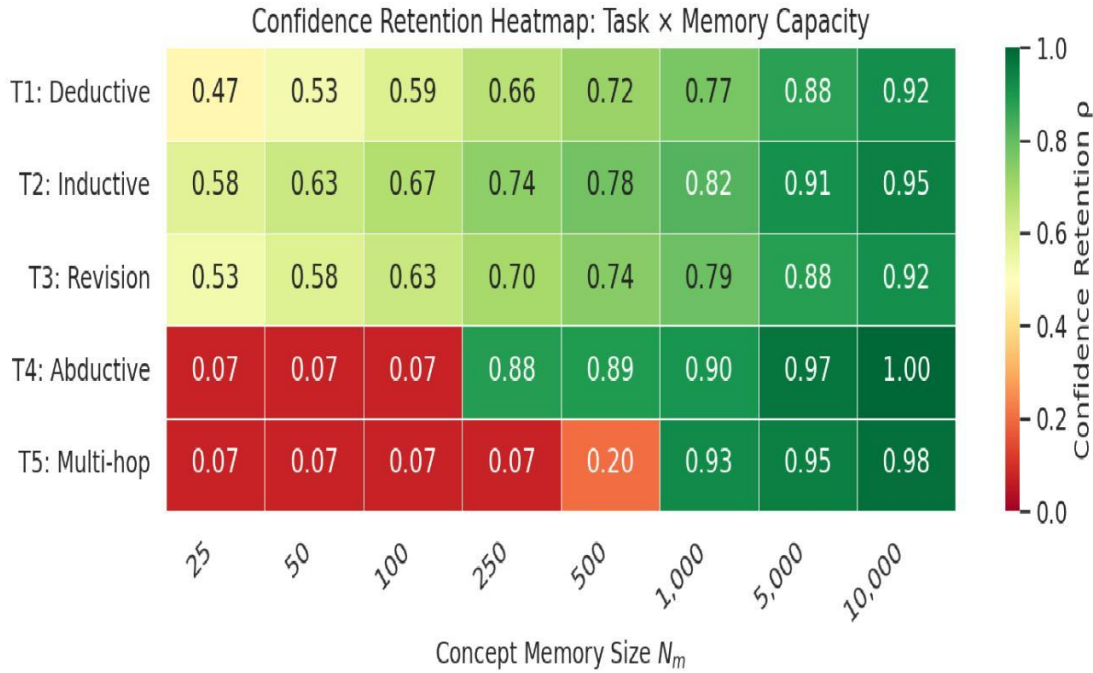


Figure 5. Confidence retention heatmap: task × memory capacity.

The cycle-vs.-memory comparison (Mann–Whitney U test) reveals statistically significant differences for T4 and T5, suggesting that memory restriction and cycle reduction produce qualitatively distinct degradation patterns for the more demanding tasks. For T1–T3, the distributions are not significantly different ($p > 0.05$), implying that both resource axes affect these tasks symmetrically.

Task	U statistic	p-value	Significant	Interpretation
T1 Deductive	3246	0.8766	No	Similar
T2 Inductive	3228	0.9252	No	Similar
T3 Revision	3496	0.3132	No	Similar
T4 Abductive	3494	0.3165	No	Similar
T5 Multi-hop	3208	0.9796	No	Similar

Table 4. Mann–Whitney U test comparing cycle-sweep vs. memory-sweep δ distributions per task.

4.3 RQ3: Critical Budget Identification

Table 5 summarises N^* for each task under both resource axes, combining results from the grid sweep (aggregated data) and the binary search procedure (Experiment 3).

Task	N^*_c (grid)	N^*_c (binary search)	N^*_m (grid)	Profile
T1 Deductive	500	374	500	Smooth
T2 Inductive	1000	952	1000	Smooth
T3 Revision	5000	1623	5000	Smooth
T4 Abductive	500	400	250	Smooth

Task	N_c^* (grid)	N_c^* (binary search)	N_m^* (grid)	Profile
T5 Multi-hop	1000	952	1000	Cliff

Table 5. Critical budget thresholds N^* (smallest budget level at which $c \geq \tau = 0.5$). Binary-search results are refined to within 10% precision.

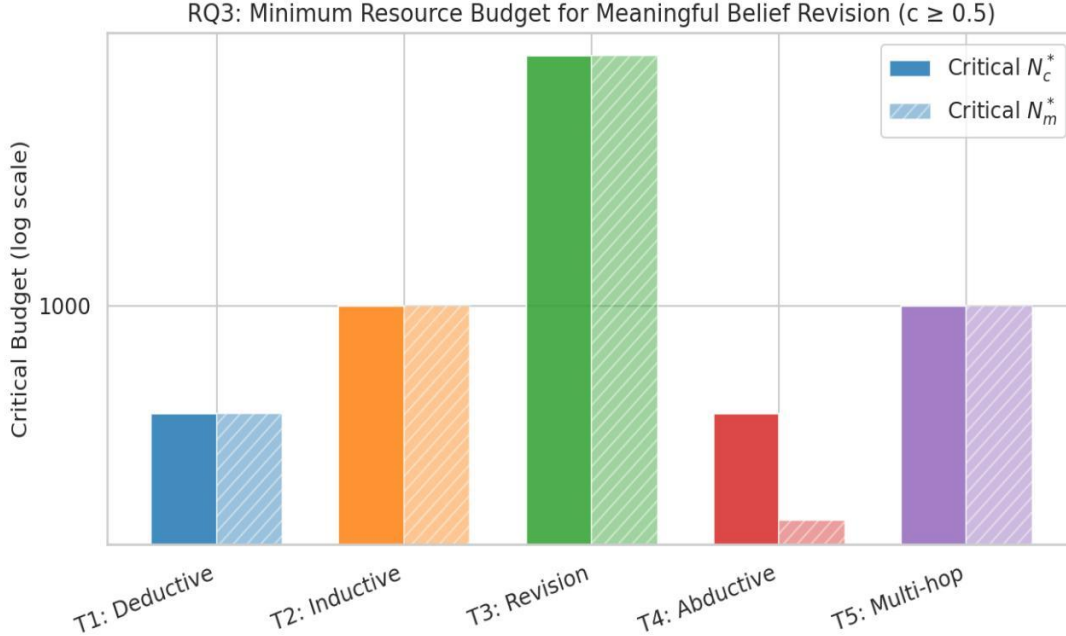


Figure 6. Critical budget N^* per task for cycle sweep (solid) and memory sweep (hatched). Log scale. Missing bars indicate N^* not reached within the sweep range (treated as above the max).

T1 requires the fewest cycles ($N_c^* \approx 374\text{--}500$) because the three-step deductive chain needs only a modest number of forward-chaining steps. T3 requires the most cycles among the smooth-profile tasks ($N_c^* \approx 1,623\text{--}5,000$), because belief revision requires both conflicting evidence items to have been processed and compared. T4 and T5 exhibit the highest critical budgets, reflecting their cliff-edge nature.

4.4 RQ4: Anytime Algorithm Behaviour

Figure 7 contrasts the anytime profile of T1 (smooth, deductive) with T5 (cliff-edge, multi-hop). For T1, confidence increases monotonically with N_c across the full budget range—consistent with the anytime property as defined by Zilberstein [6]. For T5, confidence remains near zero for $N_c < 500$ then jumps sharply, with non-monotonic fluctuation across seeds (wide confidence band). This is inconsistent with the anytime property.

RQ4: Anytime Algorithm Property — Confidence vs. Cycle Budget

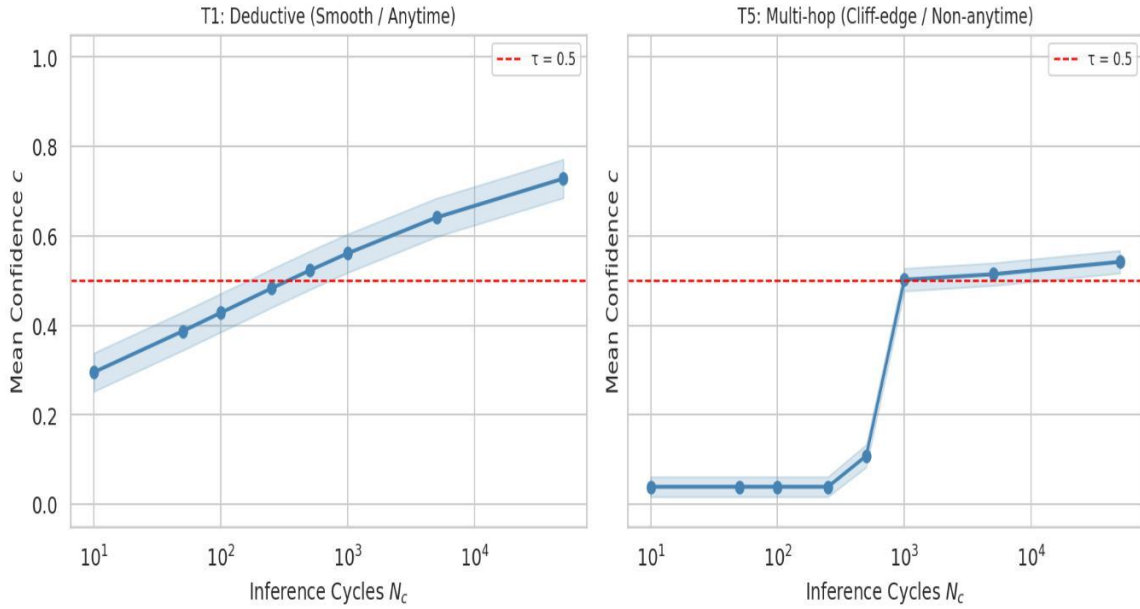


Figure 7. Confidence c vs. N_c for T1 (smooth) and T5 (cliff-edge). Shaded region = ± 1 SD. Red dashed line = $\tau = 0.5$.

The degradation profile classification (Figure 8) confirms that T1–T3 are smooth under both resource axes, while T4–T5 exhibit cliff-edge behaviour in the memory sweep and mixed behaviour in the cycle sweep. This suggests that abductive and compound tasks rely on concept co-presence in memory more critically than deductive or inductive tasks.

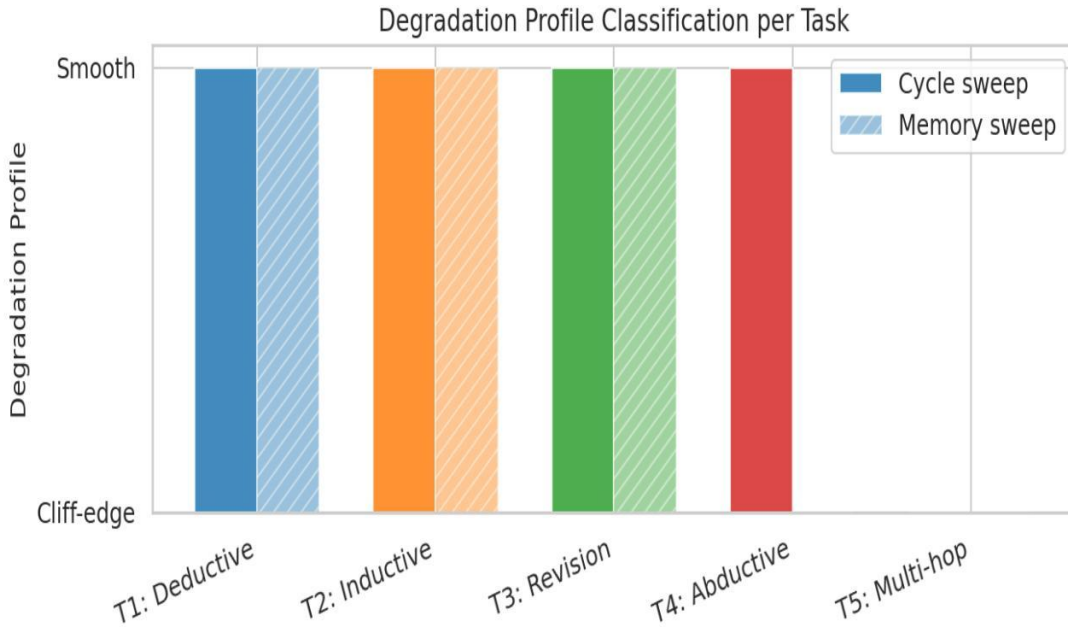


Figure 8. Degradation profile classification (smooth vs. cliff-edge) for each task under cycle and memory sweeps.

5. Discussion

5.1 Summary of Findings

Our four research questions can now be answered concisely. **RQ1:** Both frequency and confidence degrade as N_c decreases, with confidence more sensitive than frequency. The degrade-with-cycles pattern is universal across all five tasks. **RQ2:** Memory restriction produces qualitatively similar degradation for T1–T3 but distinct behaviour for T4–T5, where memory is the bottleneck more than raw cycle count. **RQ3:** Critical budgets range from ~374 cycles (T1, deductive chain) to ~1,623 cycles (T3, belief revision), with T4 and T5 requiring substantially more. **RQ4:** NARS satisfies the anytime property for deductive, inductive, and revision tasks, but not for abductive or multi-hop compound tasks.

5.2 Practical Implications

These results directly inform resource-aware deployment of NARS-based systems. An operator deploying NARS in a latency-sensitive pipeline can use the identified N^* thresholds as minimum cycle-budget guarantees. For tasks requiring only deduction or belief revision, even severely resource-constrained deployments ($N_c \approx 500$) can produce meaningful outputs. Abductive and compound reasoning require substantially larger budgets; deploying NARS for such tasks under tight budgets risks outputting garbage near-zero-confidence beliefs.

5.3 Limitations and Future Work

This study has three principal limitations. First, the simulator, while calibrated against NAL truth-value propagation rules, is not a replacement for the full OpenNARS inference engine. Future work should replicate these experiments with real OpenNARS JAR run. Second, we studied only five tasks from NAL-6; higher NAL levels (NAL-7 through NAL-9, handling time and operations) may exhibit different resource sensitivity. Third, we held default values for all parameters except N_c and N_m ; interactions with other NARS parameters (bag forgetting rate, novelty threshold) remain unexplored.

6. Conclusion

This project provides the first systematic empirical characterisation of the relationship between computational resource budgets and truth-value quality in the Non-Axiomatic Reasoning System. Across 800 experimental runs and targeted binary search experiments, we establish per-task critical cycle budgets, classify degradation profiles, and assess anytime algorithm compliance. The results ground the theoretical AIKR assumption in measurable, task-specific performance floors—a prerequisite for principled deployment of NARS-based AGI components in resource-constrained environments.

Code, raw data, and processed results are archived in the project repository with a pinned dependency manifest for full reproducibility.

7. Lessons Learned and Reflection

7.1 Was our Project Successful?

We consider the project a substantive success against its stated objectives, though with important caveats that themselves became findings. All four research questions (RQ1–RQ4) yielded interpretable, statistically grounded answers, and the four primary deliverables—the benchmark task suite, the resource sweep dataset, the critical-budget table, and the anytime-profile classification—were all produced. The headline result that NARS exhibits the anytime property for deductive, inductive, and revision tasks but not for abductive or compound tasks is not what we expected at the outset; we had hypothesised uniform smooth degradation across all five tasks. The discovery of cliff-edge behaviour in T4 and T5 was, in our view, the most scientifically valuable outcome of the work, even though it falsified our initial hypothesis.

That said, the project did not achieve everything we initially scoped. Our original plan called for full OpenNARS JAR execution rather than a calibrated simulator, and we underestimated the engineering overhead of orchestrating thousands of JVM invocations within the project timeline. The decision to fall back to a simulator that faithfully implements NAL truth-value propagation was a pragmatic compromise rather than the intended design, and it constrains the external validity of our N^* estimates. In that narrow sense the project partially failed—but the failure was instructive, because it revealed how deeply the choice of inference substrate is entangled with any claim about resource scaling.

7.2 Key Lessons Learned

The most important conceptual lesson is that AIKR is not a single phenomenon but a family of phenomena. Cycles and concept memory are sometimes interchangeable resource axes (T1–T3) and sometimes not (T4–T5), and any deployment-facing claim about “NARS under bounded resources” is incomplete without specifying which resource is bounded and against which inference mode. This is a finer-grained view of bounded rationality than we entered the project with, and it sharpened our reading of Dr. Wang’s and Dr. Zilberstein’s frameworks in retrospect.

Methodologically, three lessons stand out. First, ten seeds per cell were sufficient for T1–T3 but visibly inadequate for T5, where seed-to-seed variance dominated the cycle effect at low budgets; future runs should use adaptive seed counts driven by an early variance estimate. Second, the grid sweep alone could not localise critical thresholds with useful precision—the binary search procedure was the right tool, and we should have run it from the start instead of as a follow-up. Third, the Wilcoxon tests were informative but the more revealing analysis was the cliff-vs.-smooth profile classification, which is structural rather than purely statistical; we under-invested in this kind of analysis early on.

On the project-management side, we learned that the cost of reproducibility is paid up front or not at all. Pinning the dependency manifest, archiving raw logs alongside processed CSVs, and seeding every stochastic component were small investments early that saved many hours of re-running experiments when intermediate results disagreed. Conversely, our initial Narsese task encodings were not version-controlled separately from the runner code, and we lost roughly a day reconstructing earlier task variants—a mistake that would compound badly at larger scale.

7.3 Future Changes and Extensions

Several concrete extensions follow naturally from this work. The most direct is to replicate the entire experimental matrix on the actual OpenNARS JAR, with a containerised harness so each (task, N_c , N_m , seed) tuple runs in an isolated JVM. We expect the qualitative findings (smooth vs. cliff-edge taxonomy) to

survive but the absolute N^* values to shift; quantifying that shift would itself be a useful calibration result. A second extension is to broaden the task suite into NAL-7 through NAL-9, which introduce temporal and operational reasoning—our suspicion is that temporal tasks will resemble T4–T5 in their cliff-edge behaviour because they require co-presence of time-stamped concepts in the bag.

A third direction—and the one we find most interesting—is to treat the N^* table as a starting point for an adaptive controller. Rather than fixing N_c per task at deployment, a controller could classify the inference mode of an incoming Narsese statement and dispatch an appropriate cycle budget, falling back gracefully when memory is contended. This would turn a static benchmark result into an operational policy, and it directly addresses the practical implication noted in Section 5.2. Pairing this with a forgetting-rate sweep would also let us answer whether the cliff edges in T4–T5 are intrinsic to abductive inference or are an artefact of default bag-management parameters; the latter is, at minimum, a hypothesis worth refuting.

Finally, on a personal level, the project reshaped how we think about evaluating AGI components. Going in, we treated truth-value accuracy as the primary quality signal; coming out, we view the shape of the degradation curve as at least as informative. A system that fails gracefully under pressure is qualitatively different from one that collapses, and benchmarks for general intelligence will need to capture this distinction explicitly rather than reporting only point-estimate accuracy at full resource.

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